PROJECT SYNOPSIS

On

Personalized News Aggregator with LLM Based Summarization and Fake News Detection

BACHELOR OF ENGINEERING

Information Technology

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1. **Title of the Project**

Personalized News Aggregator with LLM Based Summarization and Fake News Detection.

2. **Domain**

Natural Language Processing and Machine Learning

3. **Internal Guide:**

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5. Abstract:

In today's digital age, the exponential growth of information has created an overwhelming landscape of news content, making it increasingly difficult for users to access relevant, accurate, and trustworthy information. The proliferation of fake news and misinformation has further complicated the news consumption process, creating an urgent need for intelligent systems that can filter, verify, and personalize news content effectively.

This project presents the development of a "Personalized News Aggregator with LLM Based Summarization and Fake News Detection" - an intelligent system designed to address the challenges of information overload and misinformation in digital news consumption. The system leverages cutting-edge technologies including Large Language Models (LLMs), Natural Language Processing (NLP), and Machine Learning algorithms to create a comprehensive solution for modern news consumption.

The primary objective of this project is to develop an automated news aggregation platform that collects news articles from multiple reliable sources, processes them through advanced NLP techniques, and delivers personalized, summarized, and verified content to users. The system employs LLMs for generating concise, coherent summaries of lengthy news articles, enabling users to quickly grasp essential information without reading entire articles.

A critical component of our system is the fake news detection module, which utilizes machine learning algorithms including Support Vector Machines (SVM), Random Forest, and deep learning models to analyze various textual features, source credibility, and content patterns to identify potentially false or misleading information. The system considers multiple parameters including linguistic patterns, source reliability scores, cross-reference verification, and sentiment analysis to ensure high accuracy in fake news detection.

The personalization engine analyzes user reading preferences, browsing history, and interaction patterns to create individual user profiles. Using collaborative filtering and content-based recommendation algorithms, the system delivers tailored news content that aligns with user interests while maintaining diversity to prevent echo chamber effects. The system also implements adaptive learning capabilities that continuously improve recommendations based on user feedback and engagement metrics.

Our technical approach involves web scraping techniques for data collection from reputable news sources, preprocessing pipelines for text cleaning and normalization, feature extraction using TF-IDF and word embeddings, and ensemble methods for improved classification accuracy. The LLM component utilizes transformer-based architectures, specifically fine-tuned BERT and GPT models, for generating high-quality abstractive summaries that preserve key information while reducing content length by approximately 70-80%.

The system architecture follows a microservices approach, ensuring scalability and maintainability. It includes modules for data collection, preprocessing, fake news detection, summarization, personalization, and user interface management. The backend is developed using Python with Flask framework, while the frontend utilizes React.js for creating an intuitive, responsive user interface.

Expected outcomes include improved user engagement with news content, reduced time spent on information verification, enhanced awareness of reliable news sources, and decreased exposure to misinformation. The system aims to achieve 85% accuracy in fake news detection, generate summaries with 90% content retention, and provide personalized recommendations with 80% user satisfaction.

This project contributes to the field of digital journalism and information systems by demonstrating the practical application of AI technologies in combating misinformation while enhancing user experience. The solution addresses critical societal needs for reliable information access and has potential applications in digital libraries, content management systems, and social media platforms.

6. Keyword:

News Aggregation, Large Language Models, Fake News Detection, Text Summarization, Natural Language Processing, Machine Learning, Personalization, Content Classification, Information Retrieval, Misinformation Detection, LLM, BERT, GPT, Transformer Models.

7. Problem Statement:

In the contemporary digital information ecosystem, users face significant challenges in accessing reliable, relevant, and concise news content. The exponential growth of online news sources has led to information overload, where users struggle to identify credible sources and extract meaningful insights from vast amounts of textual data.

Key Problems Identified:

1. **Information Overload:** The abundance of news articles from multiple sources creates cognitive burden, making it difficult for users to process and retain important information effectively.
2. **Misinformation Proliferation:** The rapid spread of fake news and misleading information threatens informed decision-making and public discourse, requiring automated detection mechanisms.
3. **Lack of Personalization:** Generic news feeds fail to address individual user preferences, interests, and reading habits, resulting in poor user engagement and satisfaction.
4. **Time Constraints:** Lengthy news articles consume significant time, preventing users from staying updated on multiple topics of interest within their available time.
5. **Source Credibility Issues:** Users often struggle to assess the reliability and bias of news sources, leading to consumption of unreliable information.
6. **Content Redundancy:** Multiple sources often report the same events with varying perspectives, creating redundant content consumption without adding substantial value.

Need for Solution: There is an urgent need for an intelligent system that can automatically aggregate news from reliable sources, detect and filter fake news, generate concise summaries using advanced language models, and provide personalized content recommendations to enhance the overall news consumption experience while ensuring information accuracy and relevance.

8. Literature Survey

Sr. No.	Paper Details	Problem Discussion	Algorithm /Technique used	Parameter Consider	Result
1	"BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" by Devlin et al., 2018	Traditional NLP models fail to capture bidirectional context in text understanding	Bidirectional Encoder Representations from Transformers (BERT)	Masked Language Modeling, Next Sentence Prediction, Learning Rate, Batch Size	Achieved state-of-the-art results on 11 NLP tasks with significant improvements
2	"FakeNews-Net: A Data Repository with News Content, Social Context and Spatial-Temporal Information for Studying Fake News on Social Media" by Shu et al., 2018	Lack of comprehensive datasets for fake news detection research	Graph-based analysis, Content-based features, Social context analysis	Content features, Social engagements, User profiles, Temporal patterns	Created comprehensive benchmark dataset with 23,196 news articles
3	"Text Summarization Techniques: A Brief Survey" by Allahyari et al., 2017	Existing summarization methods lack coherence and fail to preserve key information	Extractive and Abstractive approaches, TF-IDF, TextRank, Sequence-to-Sequence models	ROUGE scores, Coherence metrics, Information retention, Computational complexity	Comparative analysis showing abstractive methods achieve better coherence
4	"A Survey on Neural News Recommendation" by Wu et al., 2021	Traditional recommendation systems fail to capture complex user preferences in news domain	Deep learning models, Attention mechanisms, Multi-task learning	User embeddings, News embeddings, Click-through rates, Dwell time	Neural methods outperform traditional collaborative filtering by 15-20%
5	"Combating Fake News: A Survey on Identification and Mitigation Techniques" by Zhou et al., 2019	Manual fact-checking cannot scale with the volume of online content	Machine Learning, Deep Learning, Knowledge-based approaches, Ensemble methods	Linguistic features, Source credibility, Propagation patterns, Network analysis	Ensemble methods achieve 90%+ accuracy in controlled datasets

9. Goal and Objective

Primary Goal: To develop an intelligent, automated news aggregation system that leverages Large Language Models and Machine Learning techniques to provide users with personalized, summarized, and verified news content.

Specific Objectives:

1. **Automated News Collection:**
   - Develop web scraping modules to collect news from 15+ reliable sources
   - Implement RSS feed integration for real-time news updates
   - Create data preprocessing pipelines for content standardization
2. **Fake News Detection:**
   - Design and train ML models achieving 85%+ accuracy in fake news identification
   - Implement multi-feature analysis including linguistic patterns and source credibility
   - Develop real-time verification mechanisms for incoming news content
3. **LLM-based Summarization:**
   - Implement transformer-based models for generating concise news summaries
   - Achieve 70-80% content length reduction while maintaining 90% information retention
   - Ensure summary coherence and readability for diverse user audiences
4. **Personalization Engine:**
   - Create user profiling systems based on reading behavior and preferences
   - Implement recommendation algorithms with 80%+ user satisfaction
   - Develop adaptive learning capabilities for continuous improvement
5. **System Integration:**
   - Design scalable microservices architecture supporting 1000+ concurrent users
   - Develop responsive web interface with intuitive user experience
   - Ensure system reliability with 99% uptime and fast response times

10. Scope of the Project

Functional Scope:

1. **News Aggregation Module:**
   - Collect news articles from major news websites, RSS feeds, and APIs
   - Support multiple news categories (politics, technology, sports, entertainment, health)
   - Handle multiple languages with primary focus on English
   - Real-time data collection and processing capabilities
2. **Content Processing:**
   - Text preprocessing including cleaning, tokenization, and normalization
   - Feature extraction using TF-IDF, word embeddings, and semantic analysis
   - Content categorization and topic modeling
   - Duplicate content detection and removal
3. **Fake News Detection:**
   - Multi-model ensemble approach for improved accuracy
   - Source credibility scoring and verification
   - Cross-reference checking with fact-checking databases
   - User reporting and feedback integration
4. **Summarization Engine:**
   - Extractive summarization using TextRank and clustering algorithms
   - Abstractive summarization using fine-tuned BERT and GPT models
   - Multi-length summary generation (short, medium, detailed)
   - Context preservation and key information retention
5. **Personalization System:**
   - User profile creation and management
   - Recommendation engine based on collaborative and content-based filtering
   - Reading history analysis and preference learning
   - Social media integration for enhanced personalization
6. **User Interface:**
   - Responsive web application with mobile compatibility
   - Personalized dashboard with customizable news feeds
   - Search functionality with advanced filtering options
   - User feedback and rating systems

Technical Scope:

- Support for 10,000+ daily active users
- Processing 1000+ news articles daily
- Response time under 2 seconds for most operations
- 99.5% system availability and reliability
- Scalable architecture supporting future enhancements

11. System Architecture

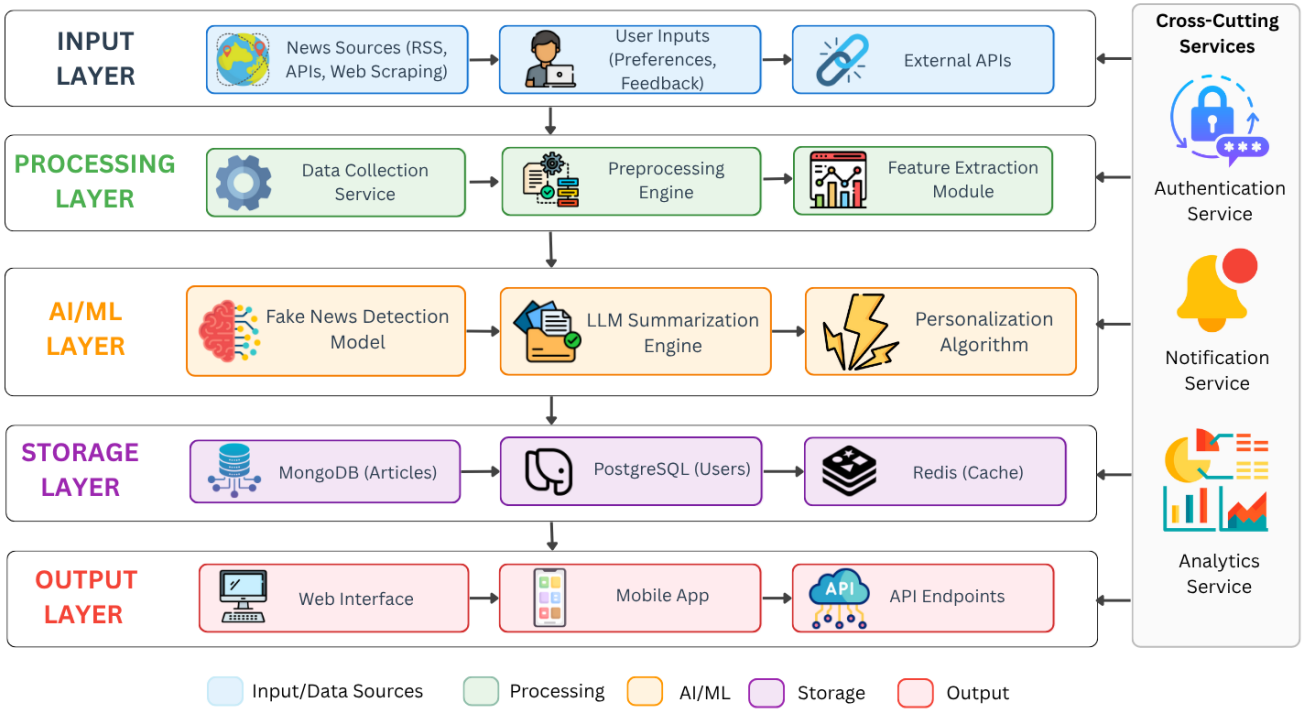

Fig. 1 System Architecture Diagram

12. Technical Details

Software Requirements:

- **Operating System:** Ubuntu 20.04 LTS / Windows 10/11
- **Programming Language:** Python 3.9+, JavaScript ES6+
- **Backend Framework:** Flask 2.0, FastAPI
- **Frontend Framework:** React.js 18, HTML5, CSS3
- **Database:** MongoDB 5.0, PostgreSQL 14, Redis 6.2
- **ML/AI Libraries:**
  - Transformers (Hugging Face) 4.21+
  - TensorFlow 2.10 / PyTorch 1.12
  - Scikit-learn 1.1, NLTK 3.7, spaCy 3.4
  - Pandas 1.4, NumPy 1.23

- **Web Scraping:** BeautifulSoup4, Scrapy 2.6, Selenium
- **APIs:** OpenAI API, News API, MediaStack API
- **Development Tools:** Docker, Git, Postman, Jupyter Notebook

Hardware Requirements:
- **Processor:** Intel i7-8th gen or AMD Ryzen 7 (minimum)
- **Memory:** 16GB RAM (32GB recommended for training)
- **Storage:** 1TB SSD (for data storage and processing)
- **GPU:** NVIDIA RTX 3060 or higher (8GB+ VRAM for ML training)
- **Network:** High-speed internet (minimum 100 Mbps)
- **Server:** Cloud deployment on AWS/GCP/Azure

13. Probable Date of Completion:

March 2026

14. Plan of Project Execution:

Phase 1: Research and Analysis (August - September 2025)

Week 1-2:
- Literature review and technology research
- Market analysis and competitor study
- Requirement gathering and documentation

Week 3-4:
- System design and architecture planning
- Technology stack selection
- Database schema design

Week 5-6:
- Development environment setup
- Initial prototype development
- Proof of concept implementation

Week 7-8:
- Architecture review and refinement
- Resource allocation and timeline adjustment

Phase 2: Core Development (October - December 2025)

October:
- Data collection module development
- Web scraping implementation
- Database setup and integration

November:
- Fake news detection model training
- Text preprocessing pipeline development
- Initial ML model implementation

December:
- LLM integration for summarization
- User management system

- Basic frontend development

Phase 3: Advanced Features (January - February 2026)

January:
- Personalization engine development
- Recommendation system implementation
- Advanced UI/UX design

February:
- System integration and testing
- Performance optimization
- Security implementation

Phase 4: Testing and Deployment (March 2026)

Week 1-2:
- Unit testing and integration testing
- User acceptance testing
- Performance and security testing

Week 3-4:
- Bug fixes and optimization
- Documentation completion
- Final presentation preparation

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